Improved NSGA2 Algorithm to Solve Multi-Objective Flexible Job Shop Scheduling Problem

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***Abstract*—The NSGA2 algorithm is one of the effective methods to solve multi-objective flexible job shop scheduling problems（MOFJSP）. An improved NSGA2 algorithm is proposed to solve the MOFJSP model that aims to minimize the maximum completion time, the total workload of all machines, the total workshop carbon emissions, the total workshop energy consumption, and the delivery time. Firstly, the improved algorithm performs neighborhood search and cross-mutation operation respectively according to the nondominated ranking level and randomly generated probability of individuals to balance their local search and global search ability of the algorithm. Then, in order to further enrich the diversity of the population and improve the solving ability of the improved algorithm, an elite retention combined with random retention is proposed to retain the parent individuals. At last, the experiment proves the effectiveness of the improved NSGA2 algorithm for solving multi-objective flexible job shop scheduling problems.**

Keywords—improved NSGA2 algorithm, multi-objective flexible job shop scheduling, elite retention strategy

I. INTRODUCTION

The flexible job shop scheduling problem (FJSP) is a kind of scheduling problem that is closest to the actual workshop scheduling. And because of this, more and more scholars are beginning to study FJSP at this stage. Reasonable solutions to such problems can not only bring more profits to the enterprise, but also effectively save resources and energy consumption and reduce carbon emissions, which is beneficial to the sustainable development of the ecological environment. Compared with the classic job shop scheduling problem (JSP), in addition to determining the processing sequence of each workpiece, FJSP also needs to allocate the machines used in each process reasonably, so FJSP is complicated NP difficult problem. In traditional FJSP, the single-objective flexible job shop scheduling problem is usually studied with the goal of minimizing the maximum completion time. In traditional FJSP, scholars mostly study the single-objective flexible job shop scheduling problem with minimizing the maximum completion time as the optimization goal. However, in actual shop scheduling, multiple goals need to be considered at the same time, so more and more scholars began to study MOFJSP. The multiple goals of MOFJSP restrict and interact with each other, so there is no optimal solution that can make multiple goals reach the best at the same time. The treatment of this problem can only be done after weighing multiple goals, as much as possible meet the needs of decision makers. Therefore, the handling of MOFJSP is more difficult than the traditional FJSP, which is a more complex NP difficult problem. In the process of solving the MOFJSP problem, intelligent algorithms have shown good solving performance, such as neighborhood search method, genetic algorithm, dispatch rule algorithm and so on.

II. RESEARCH STATUS

For MOFJSP, Guangyu Zhu et al [1] improved the genetic algorithm based on the similarity of intuitionistic fuzzy sets, and used the improved algorithm to solve MOFJSP. Haisong Huang et al [2] proposed a new discrete flower pollination algorithm for MOFJSP, which aims to minimize the maximum completion time, processing quality, production cost, and energy consumption. Lei Cao et al [3] combined the learning behavior of employees, proposed a variable neighborhood weed algorithm to solve MOFJSP; Mokhtari et al.[4] proposed the MOEA algorithm when solving the MOFJSP considering the production process and maintenance energy consumption; Carlos Soto et al [5] proposed a new parallel branch and boundary algorithm to solve MOFJSP. Chaoyong Zhang et al [6] proposed the analytic hierarchy process when solving MOFJSP with production cost, delivery delay, minimize the maximum completion time, and proposed an improved nondominated sorting genetic algorithm to carry out the established mathematical model.

The nondominated sorting genetic algorithm with elite retention strategy (NSGA2) was proposed by Srinivas and Deb on the basis of the nondominated sorting genetic algorithm (NSGA) in 2000. The NSGA2 algorithm runs faster and has good convergence performance in the process of solving. At present, NSGA2 has been applied in many fields and has achieved good results, especially when solving multi-objective problems, the algorithm has shown its excellent solution performance. However, in the process of solving the problem, there will still be problems such as insufficient population diversity and poor gool search ability .

Based on the current research status of MOFJSP and the characteristics of the NSGA2 algorithm, this paper establishes a scheduling model that aims to minimize the maximum completion time, the total workload of all machines, the total workshop carbon emissions, the total workshop energy consumption, and the delivery time. And put forward an improved NSGA2 algorithm to solve the established model.

III. MODLE BULIDING

A. Problem Description

n jobs $J= \{J_1, J_2, J_3, \dots J_n\}$ need to be processed on m different machines $M= \{M_1, M_2, M_3, \dots M_m\}$, each job has several different processes, and there are sequence constraints between different processes. Different processes of jobs can be processed on different machines with different processing times. The constraints are as follows:

*1)*At any time, one machine can only perform one processing operation.

2) At any time, the process of each job can only be processed on one machine and the process can't be interrupted.

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3) Different jobs have the same priority, and there are sequential constraints between procedures of the same job.

Problem model

The parameter description and objective function of the model are as follows:

n is the total number of jobs, m is the total number of machines, i= {1, 2, …n} is the index of job, q_i is the total number of processes of job i, j= {1, 2, … q_i } is the index of job process, k = {1, 2, …m } is the index of machine, T_i is the completion time of job i, S_{ij}: Processing time of the jth process of job i, T_{ij} is the starting processing time of the jth process of the job i, S_k: start time of machine k, T_k: finish time of machine k, U_{ij}: Processing machine number of the jth process of job i, V_k:the number of jobs processed by machine k at a certain time, P_{ijk}: Processing time of the jth process of the job i on the machine k, X_{ijk}: decision variable, indicates whether the jth process of job i is processed on machine k, X_{ijk} =1 indicates that the process j of job i is processed on machine k, and X_{ijk} =0 indicates that it is not processed on machine k, V_{tk}: decision variable, indicates that the number of jobs processed on the machine k at a certain time, C_u: carbon emission when the machine is idling, C_c: carbon emissions during machining, C_t: carbon emission during job transfer, E_u: energy consumption when the machine is idling, E_c : energy consumption during machining, E_t: energy consumption during job transfer, d_i is the delivery time of job i, a_i : penalty coefficient for delayed completion of job i .

Objective function description:

*1)Minimize the maximum completion time f*1:

$$f1 = min\ (max\ T_i)$$

*2) Minimize the total workload of all machines f*2:

$$f2 = min(\sum_{k=1}^{m}\sum_{i=1}^{n}\sum_{j=1}^{q_i} P_{ijk} X_{ijk})$$

*3) Minimize the total workshop carbon emissions f*3:

$$f3 = \min[(\sum_{i=1}^{n}\sum_{j=1}^{q_i} P_{ijk}X_{ijk})C_c + (T_k - S_k - \sum_{i=1}^{n}\sum_{j=1}^{q_i} P_{ijk} X_{ijk})C_u]$$

*4) Minimize the total workshop energy consumption f*4:

$$f4 = \min[(\sum_{i=1}^{n}\sum_{j=1}^{q_i} P_{ijk}X_{ijk})E_c + (T_k - S_k - \sum_{i=1}^{n}\sum_{j=1}^{q_i} P_{ijk} X_{ijk})E_u]$$

*5) Minimize the penalties for late delivery of jobs f*5:

$$f5 = \min\ [\ \sum_{i=1}^{n} a_i(T_i - d_i)]$$

Variable constraints: s. t

$$T_{ij} \le S_{i(j+1)} \quad (1)$$

$$\sum_{i=1}^{U_{ij}} X_{ijk} = 1 \quad (2)$$

$$V_{ijk} = \begin{cases} 1, \text{ if there is only one job on the machine } k \text{ at time } t \\ 0, \text{ otherwise} \end{cases} \quad (3)$$

$$X_{ijk} = \begin{cases} 1, \text{ if the } j^{th} \text{ process of job } i \text{ is processed on machine } k \\ 0, \text{ otherwise} \end{cases} \quad (4)$$

(1) shows that the process start time of any process of the job can't be earlier than the end time of the previous process of the job; (2) means that a process of a job can only be processed on one machine at the same time; (3) means that each machine can process only one job at a time; (4) shows whether the jth process of job i is processed on machine k, which is represented by 1 and not by 0.

IV. Improved NSGA2 Algorithm

A. Improved Individual Generation of Offspring

In this paper, the parent individuals are selected according to the individual's nondominated ranking level and the randomly generated probability, and then the neighborhood search operation and the cross mutation operation are performed respectively to effectively coordinate the search ability of the algorithm. If the parent individual satisfies the individual nondominant ranking level of 1, and the probability of random generation is less than 0.5, the neighborhood search operation is performed, otherwise the cross-mutation operation is performed.

1) The two neighborhood search methods designed in this paper are as follows:

a)Neighborhood search method 1: Randomly generate two different positions in the process gene sequence, exchange processes at these two positions to generate a new neighborhood solution.

b)Neighborhood search method 2: Randomly generate 2 different positions in the machine sequence, and replace the machines in these 2 positions with the machine with the shortest processing time among the machine set corresponding to the job process.

2) The crossover and mutation methods used in this article are as follows:

a) Crossover Operator: The crossover based on process genes uses POX, and the crossover based on machine genes uses the method of exchanging the machines corresponding to the same job process.

b) Mutation Operation: The mutation based on the process gene part adopts a random exchange order method, and the mutation based on the machine gene part adopts a random replacement method.

B. Improved Elite Retention Strategy

Aiming at the problem of insufficient population diversity in NSGA2 algorithm when solving multi-objective problems, this paper proposes an improved elite retention strategy. The specific operation process is as follows: Generate a random number between 0 and 1. If the random number is less than 0.5, select 10% of the superior individuals of the parent population to form a new parent population together with the offspring population for subsequent operations; If the random number is greater than or equal to 0.5, retain 5% of the number of parent individuals in the parent population through elite retention and random selection, respectively. And form a new parent population with offspring individuals to continue the subsequent operations.

The specific solving steps of the improved algorithm are as follows, and the flow chart is shown in Fig.1:

Step1: Population initialization, randomly generate n individuals based on the double-layer code of the job and the machine.

Step2: The greedy algorithm is used to decode the individual.

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Step3: Calculate the nondominant ranking level and crowding degree of each individual, and select parent individuals suitable for reproduction by the way of tournament selection.

Step4: According to the individual's nondominant ranking level and the probability of random generation, the neighborhood search operation and cross-mutation operation are performed on the selected parent individuals to generate offspring individuals.

Step5: The improved elite retention strategy is adopted to retain the parent individuals, and the retained individuals are merged with the offspring individuals generated by Step4 to generate a new population.

Step6: Judge whether the termination criteria is reached. If the termination criteria is reached, nondominated sorting and crowding calculation are performed on the current population, and the optimal Pareto front end is output. If not, go to step 3.

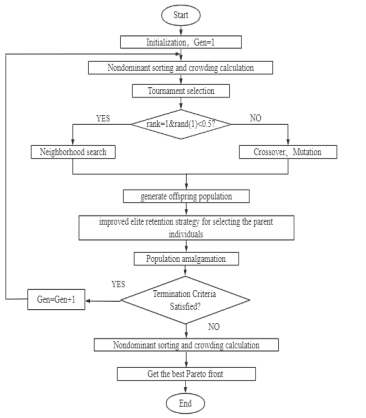


Fig.1 Improved NSGA2 algorithm flow chart

V. Simulation Experiment and Analysis

The proposed algorithm was implemented in MATLAB 2018b, the computer used is a DELL laptop with windows10 system. The hardware parameters of the computer are as follows: the CPU is Inter(R) Core (TM) i5-6200U CPU @ 2.30GHz 2.40 GHz, the memory is 4GB.

In the improved NSAG2 algorithm, for the parameter setting in the neighborhood search and the elite retention strategy, this article adopts the experimental design method to set. For the setting of different combination values, this article compares the 10 average test results of the same test data to select the most suitable combination value, and the final optimal parameter settings are as described in the text. The remaining main parameter settings are described as follows: the population size is 150, the number of iterations is 200, the crossover probability is 0.8, and the mutation probability is 0.1

In order to verify the effectiveness of this algorithm in solving multi-objective problems, minimize the maximum completion time C_m, total workload of all machines W_t and maximum machine load W_m are taken as the optimization goals, and use four KACEM [7] instances to test: 4×5 instance, 10×7 instance, 8×8 instance and10×10 instance. When solving the Pareto optimal solution, we compare the improved NSGA2 algorithm with the existing methods NBHA [8], AHGA [9], IIWO [10] and VNIWO [3]. The calculation results of KACEM instances for each algorithm are shown in Table I. Where / means no relevant data is obtained, * indicates that the set of solutions is dominated by one or more sets of solutions obtained by the improved algorithm of this article. The results of the Kacem10×10 calculation example shown in Fig.2.

TABLE I. KACEM Study Calculation Results

Example	Aims	NBHA				AHGA	IIWO			VNIWO				Improved NSGA2		
Kacem4×5	C_m	11	11	12	13	/	11	12*		/				11	12	13
	W_t	32	34	32	33	/	32	33*		/				32	32	33
	W_m	10	9	8	7	/	10	8*		/				10	8	7
Kacem8×8	C_m	14	15	16	16	14	15	15*	15*	16	17	17	18	14	15	16
	W_t	77	75	73	77	77	73	75*	75*	73	78	76	75	77	75	73
	W_m	12	12	13	11	12	12	13*	13*	13	11	12	12	12	12	13
Kacem10×7	C_m	11	12	12*		/	/			/				11	11	12
	W_t	61	60	62*		/	/			/				61	62	60
	W_m	11	12	10*		/	/			/				11	10	12
Kacem10×10	C_m	7	8	8		7	7	8	9	8*	7			7	7	8
	W_t	42	41	42		42	42	41	41	43*	42			42	43	42
	W_m	6	7	5		6	6	5	7	5*	6			6	5	5

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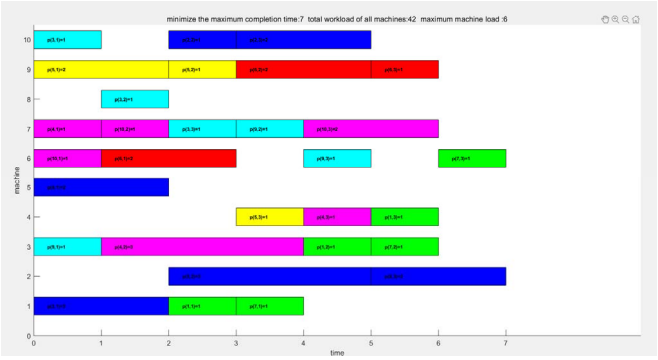


Fig.2 Gantt chart of Kacem10×10

It can be seen from the results in the table that at least one set of solutions in NBHA, IIWO, and VNIWO are dominated by the algorithm in this paper. Although AHGA does not have solutions dominated by the algorithm in this paper, the solution set sought by the algorithm in this paper is more widely distributed.

In summary, the improved algorithm in this paper has certain advantages in solving MOFJSP.

VI. Conclusion

According to the actual workshop scheduling situation, this paper establishes a scheduling model that aims to minimize the maximum completion time, total workload of all machines, total workshop carbon emissions, total workshop energy consumption, and delivery time. To better solve the established model, coordinate the search ability of the algorithm, and enhance the diversity of the population, this paper makes a series of improvements to the NSGA2 algorithm, including performing neighborhood search and cross-mutation operations according to the individual's nondominated ranking level and randomly generated probability and improving the elite retention strategy. Finally, a comparative experiment of multi-objective flexible job shop scheduling proves the feasibility and effectiveness of the improved algorithm in this paper. In the future, according to the uncertainty of flexible job shop scheduling, the dynamic multi-objective flexible job shop scheduling will be studied in depth.

REFERENCES

[1] Guangyu ZHU, Wenjie XU. Multi objective flexible job shop scheduling method for machine tool component production line considering energy consumption and quality[J]. Control and Decision, 2019, 34(02):252-260.

[2] Haisong Huang, Kai Liu, Guangyong Chu. Discrete flower pollination algorithm for multi-objective flexible workshop scheduling[J]. CIMS, 2018, 24 (11):2808-2818.

[3] Lei Cao, Chunming Ye, Xia Huang. Multi objective flexible workshop scheduling based on employee learning behavior [J]. CIMS, 2018, 24 (08):2023-2034.

[4] Mokhtari H, Hasani A. An energy-efficient multi-objective optimization for flexible job-shop scheduling problem[J]. Computers & Chemical Engineering, 2017, 104:339-352.

[5] Soto C, Dorronsoro B, Fraire H, Cruz-Reyes L, Gomez-Santillan C, Rangel N. Solving the multi-objective flexible job shop scheduling problem with a novel parallel branch and bound algorithm[J]. Swarm and Evolutionary Computation, 2020, 53:10.1016/j.swevo.2019.100632.

[6] Xiaoqiu Shi, Yanyan Li, Dingshan Deng, Wei Long. Research on FJSP based on adaptive variable level genetic weed algorithm[J]. Journal of mechanical engineering, 2019, 55(06):223-232.

[7] I. Kacem, S. Hammadi, P. Borne. Approach by localization and multiobjective evolutionary optimization for flexible job-shop scheduling problems[J]. IEEE Transactions on Systems,Man,and Cybernetics, Part C:Applications and Reviews, 2002, 32(1):1-13.

[8] M. A. F. Perez, F. M. Raupp. A newton-based heuristic algorithm for multi-objective flexible job-shop scheduling problem[J]. Journal of Intelligent Manufacturing, 2016, 27(2):409–416.

[9] Zeye Zou, Jiusun Zeng, Jinhui Cai. Improved Genetic Algorithm for Flexible Job-shop Scheduling Problem[J]. Computer Measurement & Control, 2017, 25(4):167-171.

[10] Xin Zhang, Ke Li, Dahu Yan, ZhiCheng Ji. Improved intrusion weed algorithm for solving flexible job shop scheduling problem[J]. Journal of System Simulation, 2018, 30(11):4469-4476.

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